Axles and alignment

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements and observations to improve straight rolling.

SUCCESS CRITERIA - I can...

1. Check axle parallelism and wheel clearance.
2. Record a repeatable alignment test.
3. Use measurements or observations to choose one adjustment.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **50**-turn coil has flux per turn change by **0.007 Wb** in **0.5 s**. **Find average induced-emf magnitude**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: clearance = available gap - part width.
Use positive clearance and **check free rotation**. Alignment and wheel rubbing are separate checks; change one feature at a time.

Worked reasoning: Parallel axles can still cause drag if wheels rub the chassis. Alignment and clearance are separate checks; a straight top view does not verify free rotation.

Your turn: Draw a top view and a side detail marking an axle direction and a wheel gap.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

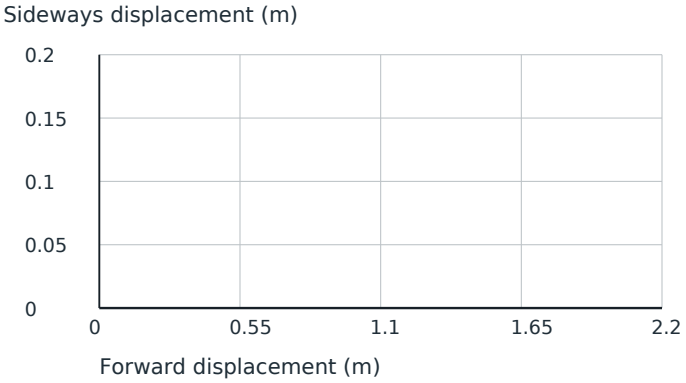
On your top-view sketch, mark both axle lines. **Add a check** for parallelism and **show clearance** between each wheel and the chassis.



Marzano check - circle: ① ② ③

04 Apply the physics

A sample straight-tracking test ends **2.0 m** forward and **0.12 m** right of the starting point. Plot the endpoint.
1. **Calculate the displacement magnitude** and direction using $\tan^{-1}(\mathbf{0.12/2.0})$.
2. **Explain what these measurements say** about straight tracking.



Marzano check - circle: ① ② ③

05 Explain your reasoning

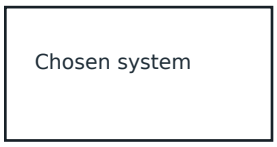
A car veers right once. Does that prove the rear axle is crooked? **Name another possible cause** and a discriminating check.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **4 Ω** resistor carries **6 A**. **Find power** using both IV and I^2R, showing the implied voltage.
Label input, useful output and other energy transfers.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

07 Exit check

Two designs each complete three identical free-roll tests.
- Before adjustment: **0.18**, **0.22**, and **0.20 m** right.
- After adjustment: **0.04**, **0.06**, and **0.05 m** right.

1. **Compare the mean offset** and **range**.
2. **Explain why these results tell** you more than one excellent run.

Trial	Before offset (m)	After offset (m)
1	0.18	0.04
2	0.22	0.06
3	0.2	0.05

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Repeated **3 m** runs finish **0.20**, **0.24**, **0.22 m** left of the line. **Find mean deviation** and range; design a controlled adjustment and a fair comparison after it.



Marzano check - circle: ① ② ③